Project: Diamond Prices

Step 1: Understanding the Model

Answer the following questions:

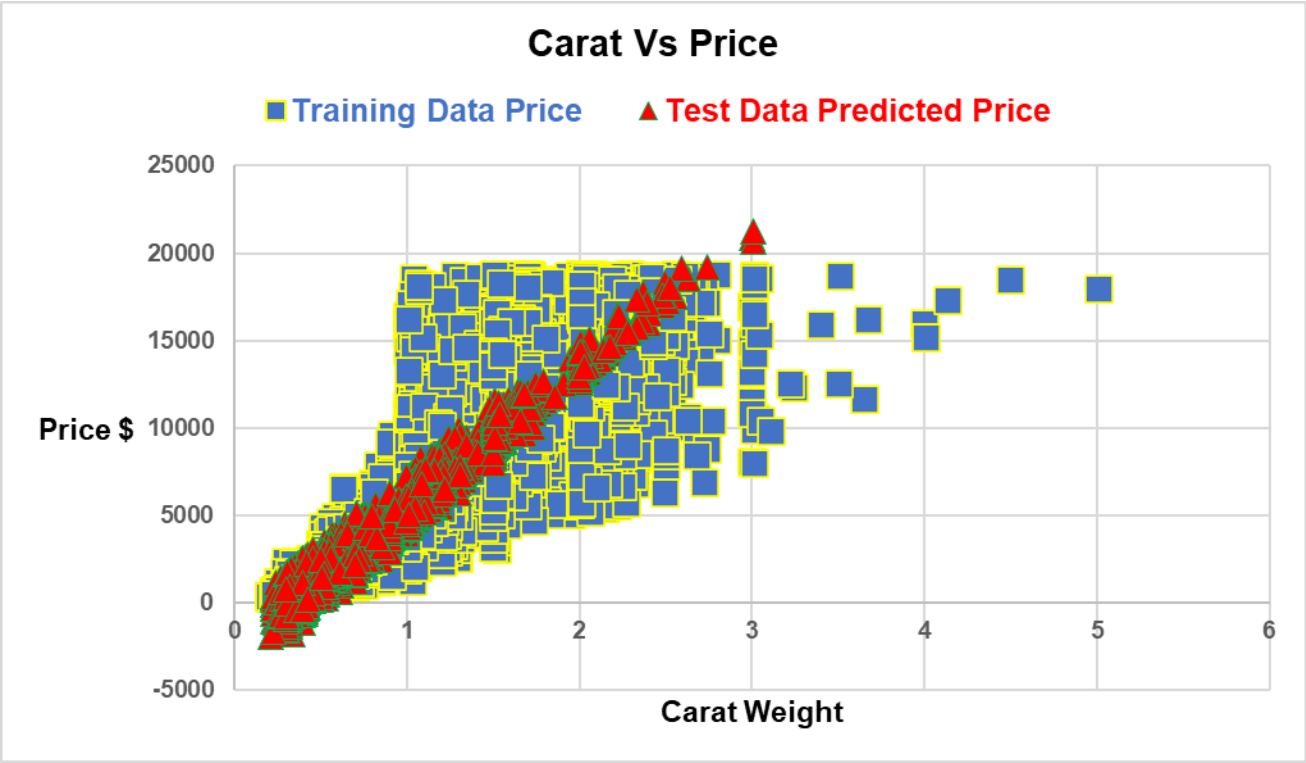
1. According to the linear model provided, if a diamond is 1 carat heavier than another with the same cut and clarity, how much more would the retail price of the heavier diamond be? Why?
    - The **one additional Carat weight** would result in an **additional $8,413 in price**.
    - The formula created by the regression is:
      [Price = -5,269 + 8,413 x Carat + 158.1 x Cut + 454 x Clarity].
    - Thus, that determined that the **coefficient for a carat** is **$8,413**, so for every **increase** in the **Carat weight** the **diamond price** will **increase** by the amount of the **coefficient**.

2. If you were interested in a **1.5 carat** diamond with a **Very Good cut** (represented by a **3** in the model) and a **VS2 clarity rating** (represented by a **5** in the model), *how much would the model predict you should pay for it?*
    - As we mentioned above, the formula created by the regression is:
      [Price = -5,269 + 8,413 x Carat + 158.1 x Cut + 454 x Clarity].
    - So, now we will plug in the values for the different variables.
        - **Predicted Price** = -5,269 + 8,413 x **1.5** + 158.1 x **3** + 454 x **5**
        - **Predicted Price** = **$10,094.80**
    - But, if we want to consider the Distributors price and cut out 30% Profit Margin, it will be:
        - **Distributors Price** = $10,094.80 * .70
        - **Distributors Price** = $**7,066.36**

Step 2: Visualize the Data

Make sure to plot and include the visualizations in this report. For example, you can create graphs in Excel and copy and paste the graphs into this Word document.

1. Plot 1 - Plot the data for the diamonds in the database, with carat on the x-axis and price on the y-axis.
2. Plot 2 - Plot the data for the diamonds for which you are predicting prices with carat on the x-axis and predicted price on the y-axis.
    - **Note**: You can also plot both sets of data on the same chart in different colors.
3. What strikes you about this comparison? After seeing this plot, do you feel confident in the model's ability to predict prices?



- **<u>Analysis, Chart observation and conclusion</u>:**

1) <u>The **predicted** prices are **more compac**t than the **actual** data</u>.

   - This is because we are not accounting for everything (attributes) that effects **diamond prices**.
   - There are many more attributes than **carat weight** that effect it. We had **Cut Quality** and **Clarity** factored into our formula **but not even that will account for all the variation**. <u>For instance</u>, this formula might look very different depending on:
     1. The **"Diamond Color"**, which is not included at mentioned Linear Equation.
     2. Any **other lurking variables** that we didn't pay attention for, such as the **"country"** we are training the model on.

2) <u>After looking at above plot</u>, the model appears **on average** to predict the Diamond prices **ok**, but it **can be very off for certain diamonds**. Even the forecast may be **very wrong** for some diamond items. <u>That are very clear at **2 situations**, as followed</u>:

A. Negative Prediction Values (about 291 diamond items)

- These **negative** values are totally **wrong** and **irrational**; thus, we **can't use** this linear equation to predict a suitable and confident price for **a specific diamond item**.
- **However**, we can use it in an aggregation form, and it would be useful and confident to some extent, such as in setting a bid price here.
- **Moreover**, if we intent to make our prediction more rational and more confident, we can deal with these **negative** values **by converting them to Zero (0)** value.
- We can see below the effect of that action, if we did so, the predicted price would be **increased** by **$136,859.70** and that would lead to **$95,801.79 increase** at recommended amount to bid.

	Predicted Price [AS IS]	Predicted Price [No Negative]	Difference
Total Predicted Price (Before Calc Profit Margin)	$ 11,733,522.76	$ 11,870,382.46	$ 136,859.70
Amount to be Bid (After Calc Profit Margin 30%)	$ 8,213,465.93	$ 8,309,267.72	$ 95,801.79

B. Different Predicted Prices at the Same Carat Weight

- At Carat weights (1, 2, and 3), as we can see the prices swings between approximately **10$ to 20,000$** for the **same carat weight**, which indicate the existence **of lurking variables**.

3) **Personally**, I think we can get more **accurate** and **trusted** result than this one, and that for below reasons:

1. Using Categorical variables as **dummy variables** can get more accurate result than converting them to ordinal variables.
2. We have **no idea about Adjusted R-Squared, P-Values**, so we can't tell, to what extent this model can **explain the variation** in the data nor **how significant** these attributes are.

4) **Finally**, While the formula may not be accurate for an individual diamond price, it should do a decent job at predicting the diamond prices that the jewelry company should pay for the 3000 diamonds price at once since it on average looks representative.

Step 3: Make a Recommendation

Answer the following questions:

1. What price do you recommend the jewelry company to bid? Please explain how you arrived at that number.
    - I **recommend** a bid of **$8,213,465.93**. I arrived at this number by using the given linear equation formula from the regression model provided that was based on **trained** diamond dataset (**50000 records**).
    - I **applied** it to the **attributes** of the **3000 new diamonds** that were up for bid.
    - I then **factored** in the **margin** the Jewelry company were looking for which was **30%** so I multiply the predicted amount **$11,733,522.76** by .**70** to get the final predicted bid of **$8,213,465.93**.